

Engineering Brief: Reframing "Beyond Linear Reasoning" for Maximum Impact

For the authors of: *Beyond Linear Reasoning: Combining Chain-of-Thought with A* Search for Robust Multi-Strategy Inference* **Purpose:** This document is a complete strategic blueprint for rewriting and reframing your paper to maximize its citation potential, novelty framing, and contribution clarity. Read every section before touching the manuscript.

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1. The Core Problem With the Current Framing

Your current paper answers "is A* complementary to CoT?" — and proves it firmly. That is a **finding**, not yet a **thesis**. The framing positions A* as a positive tool that sometimes helps. The far more publishable and citable framing is the **inverse**: A*, the theoretically optimal cost-based search algorithm, **systematically fails in specific, predictable conditions**, and those conditions reveal something deep about how LLMs reason.

This is not just a stylistic reframe. It changes the entire contribution:

Current Frame	Proposed Frame
"A* and CoT are complementary"	"Even the best algorithm fails without a model that can execute it faithfully"
A* is presented as a positive tool	A* is used as a diagnostic lens to expose <i>LLM reasoning structure</i>

Routing is the main challenge	The interaction between search topology and model capability is the main challenge
Weak SLM results are a footnote	SLM results become a new research axis (search helps most where models are weakest)
Failure modes are in an appendix	Failure modes are foregrounded as the primary contribution

The proposed title captures this shift: **"The Best Is Not Always Best: When A* Search Fails to Improve LLM Reasoning and Why"**

Subtitle option: *A Diagnostic Study of Reasoning Topology, Model Capability, and the Limits of Cost-Based Search*

2. Closest Related Papers

Here is a curated, honest comparison of every paper that does something similar and where you stand relative to each.

2.1 Tree of Thoughts (ToT) — Yao et al., NeurIPS 2023

What they did: Introduced tree-structured reasoning for LLMs using BFS/DFS over "thoughts." Showed large gains on Game of 24 (4% CoT → 74% ToT with GPT-4). Established that branching improves hard planning tasks.

Where you beat them:

- ToT uses **undirected BFS/DFS** — it has no principled cost function. Your A* adds an explicit, theoretically-motivated path cost (Equation 5) and a conservative admissibility proof.
- ToT does **not compare against CoT on natural language QA benchmarks** like StrategyQA or HotpotQA — you do.
- **Critically:** ToT does not ask *when* search hurts. You do. This is your strongest differentiator.
- You compare ToT directly in Table 1 and show A* beats it on all three benchmarks with the same base model.
- You test on a **weaker base model** (LLaMA-3.1-8B), making your results more reproducible and accessible.

Citation opportunity: Cite ToT and explicitly frame your admissibility theorem as fixing a gap ToT left open ("ToT lacks a principled priority rule; we address this with a conservative

admissible heuristic under Theorem 1").

2.2 Reasoning with World Model (RAP) — Hao et al., EMNLP 2023

What they did: Used MCTS over reasoning traces where the LLM acts as both a world model and a policy. Showed improvements on planning and question answering tasks.

Where you beat them:

- MCTS requires rollout simulations to estimate node value, which is **computationally heavier** than your heuristic-guided A*. Your system is budgeted and tractable.
 - They do not provide a **complementarity analysis** — they don't show when their method fails and CoT wins.
 - Your failure mode taxonomy (overthinking, spurious association, step repetition) is something their paper never attempts.
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2.3 Self-Evaluation Guided Beam Search — Xie et al., NeurIPS 2024

What they did: Applied beam search to reasoning traces where the model scores each step using self-evaluation. Showed improvements on math and coding tasks.

Where you beat them:

- Beam search is undirected in cost space — it keeps the K highest-scored nodes regardless of actual trace length or uncertainty. Your A* path cost is more principled.
 - They did not analyze **failure modes or routing**.
 - No complementarity analysis. No SLM vs. LLM breakdown.
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2.4 Chain-of-Thought Prompting — Wei et al., NeurIPS 2022 (the original)

What they did: Established that intermediate reasoning steps improve LLM performance.

Your contribution relative to them:

- You show that CoT's linear generation **creates systematic failure modes** (early commitment, no backtracking).
 - You quantify exactly how often CoT fails where A* succeeds and vice versa (Table 2 — the complementarity decomposition).
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2.5 Self-Consistency — Wang et al., ICLR 2023

What they did: Sample multiple CoT paths, take majority vote. Simple, effective, no search needed.

Where you beat them:

- Table 1 directly shows A* beats self-consistency on StrategyQA (74.80% vs. 71.22%) and HotpotQA (80.10% vs. 78.21%).
 - Self-consistency allocates extra compute **blindly** — all paths are equally likely. A* allocates it **guided by a heuristic**.
 - The comparison of "how to spend extra inference budget" is your most policy-relevant finding and should be elevated.
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2.6 Coconut (Chain of Continuous Thought) — Hao et al., arXiv Dec 2024

What they did: Instead of generating text tokens, they feed the last hidden state back as continuous input, enabling implicit BFS-style search. Outperforms CoT on logical reasoning tasks.

Your relevance: Your work is the discrete, interpretable counterpart. Coconut's search happens invisibly inside the model. Your A* search happens transparently in natural language. This is a **philosophical differentiator** — you produce **auditable, human-inspectable** reasoning traces. Frame this explicitly.

2.7 "Don't Get Lost in the Trees" (Fetch Framework) — arXiv 2025

What they did: Identified over-exploration (redundant states) and under-exploration (high-variance verifier scoring) as key inefficiencies in tree search for LLMs. Proposed semantic clustering to merge similar nodes.

Your connection: Their identified failure modes (redundant states, verifier instability) correspond directly to your observed A* failure modes (step repetition, spurious association). **You are providing the same diagnosis from a different angle** — they approach it as an efficiency problem, you approach it as a reasoning correctness problem. Cite each other's framing. This strengthens both papers.

2.8 rStar-Math — Guan et al., 2025

What they did: Used MCTS with small language models (SLMs) to achieve near-o1 performance on math reasoning by generating verified step-by-step CoT trajectories.

Your connection — VERY IMPORTANT: This paper establishes that **search dramatically helps SLMs**. Your Qwen-0.5B results (HotpotQA: 8.40% CoT → 30.00% A*, a 3.5x

improvement) are the same finding in a QA setting. You must explicitly cite rStar-Math and position your SLM result as QA-domain confirmation of their math-domain result. This gives your SLM finding a prestigious anchor.

2.9 Stop Overthinking Survey — arXiv 2025

What they did: Comprehensive survey showing reasoning models (o1, DeepSeek-R1) suffer from "overthinking" — generating unnecessarily long chains that sometimes hurt accuracy.

Your connection: Your A* failure mode #1 is **literally overthinking**: the model overcomplicates simple problems. You empirically demonstrate this in a controllable setting (Appendix A.3). You should cite this survey and position your failure taxonomy as **the first controlled instance-level study of overthinking in search-based reasoning**, not just in long-CoT reasoning models.

2.10 Decreasing Value of CoT — Wharton Report, 2025

What they did: Showed that CoT prompting adds little value for reasoning-native models (o1, o4-mini), and that non-reasoning models gain only modestly and inconsistently.

Your connection: Their finding that "CoT is not universally beneficial" is the **same claim** you are making for A*. You are the search-side version of their prompting-side finding. This is very citable cross-pollination — your paper and theirs form a coherent message: neither CoT nor search-based methods are universally optimal. The real challenge is selecting the right paradigm per task.

3. Where Your Paper Clearly Wins

A consolidated list of your strongest empirical and theoretical claims, ranked by impact:

Rank 1 — The Oracle Gap Is Enormous and Nobody Has Characterized It Like This

- Oracle margins: +9.79 (StrategyQA), +13.51 (LogicQA), +14.10 (HotpotQA)
- This is stronger than anything ToT, RAP, or self-consistency papers reported in terms of **headroom evidence**
- **No prior paper** has explicitly shown that the bottleneck is *strategy selection*, not search quality itself
- This is your most citable finding — it reframes the entire test-time compute debate

Rank 2 — The SLM Finding Is a New Research Direction

- Qwen-0.5B: CoT → A* on HotpotQA: 8.40% → 30.00% (+21.6 absolute points, 3.5x

relative)

- On StrategyQA: essentially tied (53.60% vs. 53.40%) — search neither helps nor hurts
- On LogicQA: A* actually **hurts** SLMs (24.00% → 20.20%) — search degrades performance when the base model cannot generate reliable successor steps
- **Interpretation:** A* acts as a capability amplifier for SLMs only on multi-hop tasks where evidence is distributed and the base model has *some* factual coverage. When the task requires compact logical coherence (LogicQA), the base model generates bad successors and search compounds the error.
- **This is a novel finding about the interaction between model capability and search topology** — nobody has characterized it this way

Rank 3 — A Formal Admissibility Theorem for Natural Language Reasoning

- Prior work (ToT, MCTS-based methods) uses heuristics without formal guarantees
- Your Theorem 1 is the first (to our knowledge) admissibility result for a heuristic over *natural language reasoning traces*
- This is a theoretical contribution the community will cite even if they don't use your exact method

Rank 4 — The Failure Mode Taxonomy

- Three failure modes: overthinking simple problems, spurious associative expansion, budget waste via repetition
- These are empirically grounded (Appendix A.3), not speculative
- They align with the broader "overthinking" literature, making them cross-citable
- **No prior search-based LLM reasoning paper** has characterized failure modes this systematically

Rank 5 — Routing as a Research Problem in Its Own Right

- Best router (random forest): 76.25% vs oracle 84.59% on StrategyQA — a 8.34 point gap remains
- You demonstrate that even an LLM-as-critic (with access to both traces) cannot close the gap
- This positions **routing as an open problem**, which is highly citable as future work motivation

4. Proposed New Title and Reframing Philosophy

Primary Title

"The Best Is Not Always Best: When A* Search Fails to Improve LLM Reasoning and Why"

Why This Title Works

- Creates cognitive dissonance (readers know A* is optimal — why would it fail?)
- Directly inverts the expected narrative, making it memorable
- "and Why" signals a causal explanation, not just an empirical catalog — reviewers love causal arguments
- It is honest: A* does fail, often, and you have characterized it

Alternative Titles (order by preference)

1. **"Optimal Search, Suboptimal Reasoning: A Diagnostic Study of A* Over LLM Reasoning Traces"**
2. **"Reasoning Topology Matters More Than the Algorithm: A* Search, Chain-of-Thought, and the Strategy Selection Gap"**
3. **"From Oracle to Router: Characterizing the Gap Between Complementary Reasoning Strategies in LLMs"**

Subtitle (keep one)

- *A Diagnostic Study of Reasoning Topology, Model Capability, and the Limits of Cost-Based Search*
- *Or: Why the Bottleneck Is Not Search Itself, But Knowing When to Search*

The Reframing Philosophy in One Sentence

The paper should read as: **"We used A* as a scalpel to dissect how LLMs reason, and what we found is more interesting than A* itself."**

5. The New Narrative Arc

The paper should be restructured around this spine:

Act 1 — Setup (Introduction): The Optimality Paradox

Open with the paradox directly: A* is the gold standard of search — it is theoretically optimal under admissible heuristics, cost-complete, and used everywhere from GPS routing to game AI. So why does it fail to improve reasoning in LLMs over 35% of the time? And why does a simple linear text generator (CoT) sometimes outperform it? This contradiction is the puzzle the paper resolves.

Do NOT open with "CoT is widely used but has limitations." Every paper does this. Open with

Do NOT open with "CoT is widely used but has limitations." Every paper does this. Open with the paradox.

Opening sentence (draft):

"A* search is provably optimal under admissible heuristics — yet on LogicQA, applying it to LLM reasoning traces *reduces* accuracy compared to a single left-to-right chain of thought. We investigate why."

Act 2 — Empirical Foundation: The Complementarity Fact

Present Table 1 and Table 2 early and clearly. The oracle gap is the central empirical fact. Frame it as: "The problem is not that A* or CoT is weak. The problem is that neither is dominant, and we can prove it with exact instance-level counts."

Act 3 — The Diagnosis: What A* Actually Does to Reasoning

This is the new centerpiece. Reorganize your qualitative and failure-mode analysis into a Reasoning Topology Framework with four named categories:

Topology Type	A* Outcome	CoT Outcome	Why
Non-linear evidence aggregation	✓ Wins	✗ Loses	Deferred commitment helps
Hypothesis elimination (multi-choice)	✓ Wins	✗ Loses	Parallel branch comparison helps
Multi-hop chaining	✓ Wins	✗ Loses	Bridge entity search helps
Direct factual recall	✗ Loses	✓ Wins	Single path is sufficient; branching introduces drift
Globally coherent argumentation	✗ Loses	✓ Wins	Fragmentation breaks logical frame

This table should be in the paper body, not the appendix. It is your most actionable contribution for practitioners.

Act 4 — The SLM Axis: A New Variable Nobody Considered

Dedicate a full subsection to the SLM results. Frame it as: "Search is most valuable not just for hard tasks, but for *underpowered models on hard tasks*." The 3.5x gain on HotpotQA for Qwen-0.5B tells a story about search compensating for limited parametric knowledge — but only when the model has enough capacity to generate non-trivially different successors.

Act 5 — Why Routing Is Unsolved: The Fluency-Correctness Asymmetry

Your observation that the LLM-as-critic prefers CoT traces because they are "more fluent" even when they are logically weaker is a **crucial insight** that needs its own named concept. Call it the **Fluency-Correctness Asymmetry**: in natural language reasoning, the trace that reads better is not necessarily the trace that reasons better. This is why routing is hard, and it explains the gap between critic-based routing and the oracle.

This single insight is citable in its own right.

Act 6 — Conclusion: The Open Problem Statement

End with a crisp statement of the open problem: "The bottleneck is not building a better search algorithm — it is building a router that understands reasoning topology. We have characterized the topology. Building the router is the next frontier."

6. New Sections and Findings to Add

These are specific additions the writer should incorporate based on the experimental data you described having:

6.1 Add: A Failure Rate Analysis Table

You mentioned having experiment logs. Create a table like this from your data:

Failure Mode	Frequency (A* wrong, CoT right)	Example Dataset	Description
Overthinking Simple Problems	~X% of A*-only-wrong	StrategyQA	Overcomplicates a 2-fact recall
Spurious Associative Chain	~X% of A*-only-wrong	StrategyQA	Weak semantic link becomes an inference chain
Step Repetition / Budget Waste	~X% of A*-only-wrong	StrategyQA	Same node paraphrased across steps
Fluency Trap (critic error)	~X% of critic-routing failures	All datasets	Critic picks CoT because it's more readable

Even rough percentages strengthen the paper enormously. Fill in your actual numbers.

6.2 Add: A Model Capability × Task Topology Interaction Analysis

The key insight is: **A* gain = f(model capability, task topology)**. Create a 2×2 or 2×3 matrix:

Model	Task Topology	
	Simple/Linear	Complex/Branching
	A*≈CoT	A*>>CoT

Your Qwen-0.5B results give you the "Weak" row data. Your LLaMA-3.1-8B results give you the "Strong" row. This is a genuinely novel interaction effect and should be tested and reported as such.

6.3 Add: A Token Cost Analysis

A* uses more tokens. You should report the average token count per strategy and compute an accuracy-per-token metric:

Accuracy-per-1K-tokens: CoT vs A* vs Self-Consistency vs ToT

This is critical for practitioners and reviewers who will ask "is the compute cost justified?" You have partially addressed this in the FAQ appendix — bring it to the main body.

6.4 Add: A Confidence Calibration Analysis

Your heuristic uses self-reported confidence scores. A key limitation is that LLMs are poorly calibrated. Add a short analysis: how correlated is self-reported confidence with step correctness in your traces? Even a simple scatter plot or binned accuracy-by-confidence analysis will strengthen your admissibility claim and be honest about its practical limits.

7. SLM Finding — The Hidden Gem You Must Amplify

This deserves its own section because it is the most practically important and most underexploited result in the paper.

What the data shows:

Model	Dataset	CoT	A*	Gain	Interpretation
Qwen-0.5B	HotpotQA	8.40%	30.00%	+21.6 pts (3.5x)	Search amplifies SLM on multi-hop
Qwen-0.5B	StrategyQA	53.60%	53.40%	~0	Search neutral on commonsense
Qwen-0.5B	LogicQA	24.00%	20.20%	-3.8 pts	Search damages SLM on compact logic
LLaMA-3.1-8B	HotpotQA	76.80%	80.10%	+3.3 pts	Smaller but consistent gain
LLaMA-3.1-8B	StrategyQA	64.52%	74.80%	+10.28 pts	Large gain on commonsense
LLaMA-3.1-	LogicQA	45.78%	44.85%	-0.93 pts	Slight loss on logic

The pattern:

- On **HotpotQA** (multi-hop, distributed evidence): A* helps ALL models, but helps SLMs **dramatically more** in relative terms.
- On **LogicQA** (compact logic): A* hurts or is neutral for BOTH model sizes — this is a task topology effect, not a model size effect.
- On **StrategyQA** (commonsense): A* helps the strong model a lot but is neutral for the weak model — this is a capability threshold effect.

Why this matters:

The SLM result suggests that search compensates for **limited parametric knowledge** — when the model doesn't have facts memorized, exploring multiple branches increases the chance of retrieving the right one. But this only works when:

1. The task requires finding distributed evidence (HotpotQA style), not compact inference (LogicQA style)
2. The model is capable enough to generate meaningfully different successors (Qwen-0.5B can, barely, for factual QA)

What to write:

Frame a dedicated subsection: "**Search as Capability Amplification for Small Models**" with the hypothesis: *"For underpowered models on evidence-distributed tasks, A* search partially compensates for limited parametric coverage by exploring alternative retrieval paths."* Test this hypothesis with your data. This is a publishable sub-finding.

8. Failure Mode Taxonomy

Your Appendix A.3 failure modes are gold. They should be in the main paper body, named, and connected to the broader literature. Here is how to present them:

Failure Mode 1: The Overthinking Trap

What happens: When the answer requires direct factual recall, A* creates unnecessary semantic branches. The "Is Disney's ice princess real?" example — A* correctly identifies Elsa but then argues over "queen" vs. "princess" and gets it wrong.

Why it happens: The LLM's confidence scoring doesn't distinguish between "this step is factually decisive" and "this step is analytically interesting." Both receive similar confidence signals, so the search continues expanding after the answer is already reachable.

Connection to literature: This is the search analog of the "overthinking problem" documented in long-CoT reasoning models (Stop Overthinking survey, 2025). You should make this connection explicit.

Proposed fix for future work: A **sufficiency detector** — a module that flags when the current partial trace is already sufficient to answer with high confidence, terminating the search early.

Failure Mode 2: Spurious Associative Expansion

What happens: One search branch follows a weak semantic association (menthol → cold weather → Thanksgiving) and the heuristic keeps it alive because it mentions relevant concepts without actually reasoning correctly.

Why it happens: Your entity-coverage heuristic rewards traces that mention task-relevant concepts, even if the logical connection is spurious. The heuristic cannot distinguish causal from associative relevance.

Connection to literature: The "Fetch" paper (2025) identifies this as "over-exploration due to semantically similar but logically distinct states." They propose clustering; you found the same pathology first in NL reasoning.

Proposed fix: A **logical entailment filter** at the successor generation stage — require each step to logically follow from the previous, not just share concepts.

Failure Mode 3: Budget Waste via Step Repetition

What happens: The search generates near-duplicate steps that exhaust the node budget without exploring new territory.

Why it happens: The successor generation function has no memory of previously expanded states (beyond what's in the trace), so semantically identical steps get regenerated with different surface forms.

Connection to literature: ToT and MCTS papers noted this; Fetch proposes deduplication via SimCSE clustering. Your contribution: you show this is **quantitatively significant** in natural language QA settings.

Proposed fix: Signature-based deduplication using sentence embeddings before adding to the OPEN queue.

9. Instructions to the Writer

This section is a direct, actionable checklist for rewriting the paper. Work through it in order.

Step 1: Rewrite the Abstract (Critical)

The abstract must lead with the paradox, not the method. Draft:

"A* search is theoretically optimal under admissible heuristics — yet applying it to LLM reasoning traces fails to improve, and sometimes damages, performance in over 35% of cases. We study *why*, using a budgeted A*-style search procedure alongside standard Chain-of-Thought (CoT) prompting across StrategyQA, LogicQA, and HotpotQA. Our main findings are: (1) CoT and A* exhibit strong instance-level complementarity, with oracle gaps of +9.79, +13.51, and +14.10 points above the stronger single method; (2) A* fails systematically on direct-recall and globally-coherent reasoning tasks via three identifiable failure modes; (3) search dramatically amplifies small language model (SLM) performance on multi-hop tasks (+21.6 points for Qwen-0.5B on HotpotQA) but degrades it on compact-logic tasks; (4) the central bottleneck is not search quality, but *strategy selection* — current routers recover only a fraction of the oracle gain. We introduce a Reasoning Topology Framework to characterize when each strategy wins and provide a formal admissibility proof for the heuristic. Our results position strategy selection, not search design, as the core open problem in LLM inference-time reasoning."

Step 2: Restructure the Introduction

Remove the "CoT has limitations" opening. Replace with:

1. **The Paradox** (A* is optimal; why does it fail?)
2. **The Stakes** (test-time compute is exploding; we need principled allocation)
3. **The Approach** (use A* not to replace CoT but to diagnose when search helps vs. hurts)
4. **The Contributions** (cleaner list — see below)

Revised contributions:

1. A budgeted A*-style reasoning procedure with a formal conservative admissibility theorem
 2. The first instance-level complementarity decomposition between search-based and linear LLM reasoning across three benchmarks, revealing a large oracle gap
 3. A Reasoning Topology Framework mapping task structure to optimal reasoning strategy
 4. Evidence that A* acts as a capability amplifier for SLMs on evidence-distributed tasks but degrades performance on compact-logic tasks
 5. Three empirically-grounded failure modes of A* search in NL reasoning, with proposed mitigations
 6. A characterization of the Fluency-Correctness Asymmetry that explains why LLM-based routing is hard
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Step 3: Add the Reasoning Topology Framework (New Section 3.x)

Build a clean table and discussion as described in Section 5 of this brief. Use the five topology types. This table is the most practitioner-citable element in the paper.

Step 4: Move Failure Modes to Main Body

Take Appendix A.3 and expand it into a main-body section titled "**Systematic Failure Modes of A* in Natural Language Reasoning**". Add the connection to the overthinking literature (cite the 2025 survey). Add the fluency-correctness asymmetry as a fourth failure mode at the routing level.

Step 5: Add the SLM Dedicated Subsection

Write "**Search as Capability Amplification for Small Language Models**" in Section 3. Include the table from Section 7 of this brief. State the hypothesis clearly. Discuss the capability threshold (why Qwen-0.5B gains on HotpotQA but loses on LogicQA). Connect to rStar-Math (Guan et al., 2025).

Step 6: Add the Token Cost Table

Report average tokens-per-query for each method. Compute accuracy-per-1K-tokens. This goes in Section 3 or an efficiency subsection. Even a 2-row, 4-column table will satisfy reviewer concerns.

Step 7: Update the Related Work

Add citations and discussion for:

- **Coconut** (Hao et al., 2024) — latent-space alternative to discrete search
 - **Fetch / Don't Get Lost in the Trees** (2025) — over/under-exploration in tree search
 - **rStar-Math** (Guan et al., 2025) — MCTS for SLMs in math
 - **Stop Overthinking Survey** (2025) — connects your failure mode 1 to the broader literature
 - **Wharton CoT Report** (Meincke et al., 2025) — your search-side mirror in the prompting world
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Step 8: Add a Limitations Subsection to Discussion (Not Just at the End)

Frame limitations as *research opportunities*, not apologies:

- Confidence miscalibration → Future: calibrated uncertainty in search
 - Feature-engineered router → Future: topology-predictive routing
 - Single model family → Future: test on reasoning-native models (o1, DeepSeek-R1)
 - No fine-tuning → Future: jointly train routing and search policies
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Step 9: Rewrite the Conclusion

The conclusion should be punchy and declarative. End with the open problem statement. Avoid listing what you did — the reader just read it. Instead, state what the field should do next.

Draft final paragraph:

"The right question is no longer whether search can improve LLM reasoning — it can, and it does, in specific, characterizable conditions. The right question is how to recognize those conditions at inference time. Building a router that understands reasoning topology — that can distinguish a problem requiring non-linear evidence aggregation from one requiring a single coherent argumentative frame — is the next core challenge in test-time compute. We have provided the diagnostic framework. The solution remains open."

Step 10: Code and Reproducibility

Your current draft links to code. Make the README clearly describe:

- How to reproduce the SLM (Qwen-0.5B) experiments
- The failure mode classification script (if you have one)
- The routing strategies and their feature sets

This improves citation likelihood by making your results usable.

10. Potential Impact Analysis

Here is an honest, hard-nosed assessment of where this paper can land and what it needs to get there.

Current Ceiling: Strong Workshop / Short Paper at ACL/EMNLP

With the current framing, this paper reads as an empirical comparison study with moderate novelty. The theoretical contribution (Theorem 1) is real but scoped narrowly. The results are genuine but the "A* + CoT are complementary" finding is incremental given ToT and MCTS

papers.

Potential Ceiling with Reframe: Main Conference ACL / EMNLP / NAACL (Long Paper)

With the proposed reframe, the paper makes several distinct contributions that reviewers can point to:

- The oracle gap characterization (novel, quantified)
- The reasoning topology framework (practical, novel)
- The SLM amplification finding (novel sub-finding, connects to rStar-Math)
- The failure mode taxonomy (actionable, empirically grounded)
- The fluency-correctness asymmetry (conceptually novel, explains routing difficulty)

What Would Push It to ICLR / NeurIPS Territory

To reach the top venues, you would need one of:

1. A trained routing model that significantly closes the oracle gap (even +3-4 points above the best current router)
2. An extension to reasoning-native models (o1-mini, DeepSeek-R1-Distill) — showing that even these models benefit from external A* routing at inference time
3. A fine-tuning experiment where A*-collected traces are used to distill routing capability into a smaller model (connecting to the CPO / DPO literature)

Citation Potential by Finding

Finding	Expected Citations in 2 Years
Oracle gap + complementarity decomposition	High (20-40) — referenced in every routing/test-time compute paper
Reasoning topology framework	High (20-35) — referenced in practical "which method when" guides
Admissibility theorem	Medium (10-15) — specialized theory citations
SLM amplification finding	Medium-High (15-25) — connects to growing SLM literature
Failure mode taxonomy	High (25-40) — everyone who builds search systems will cite this
Fluency-correctness asymmetry	Medium (10-20) — cited in LLM evaluation and routing papers

The One Thing That Will Make or Break Publication

The reviewers will ask: **"What is the new insight that changes how I think about test-time compute?"**

Your answer must be: **"The bottleneck is not search quality — it is reasoning topology recognition. We prove this with a quantified oracle gap and explain it with a failure mode taxonomy that the community can build on."**

If every section of the rewritten paper supports that answer, the paper is publishable at a main venue.

Appendix: Suggested New Citation List to Add

Paper	Venue	Why Cite
Hao et al. (2024) — Coconut	arXiv/NeurIPS	Latent-space alternative to your explicit search
Zhang et al. (2025) — Fetch	arXiv	Same failure mode identification from efficiency angle
Guan et al. (2025) — rStar-Math	arXiv	SLM + search = large gains; your QA equivalent
Meincke et al. (2025) — Decreasing Value of CoT	SSRN	Your search-side mirror; CoT also context-dependent
Chen et al. (2024) — Overthinking of o1	arXiv	Connects your failure mode 1 to the wider community
Zhang et al. (2025) — Stop Overthinking Survey	arXiv	Broad framing for your failure mode analysis
ToTRL (2025)	arXiv	RL-trained ToT — shows ToT itself is being improved; positions your analysis of ToT limits

Document prepared April 2026. All benchmark numbers and claims reference the uploaded paper manuscript.